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on

Occlusion Removal and Inpainting of Crowdsource Images towards 3D Reconstruction of Heritage Sites

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CERTIFICATE

This is to certify that project entitled **"Occlusion Removal and Inpainting of Crowd-source Images towards 3D Reconstruction of Heritage Sites"** is a bonafide work carried out by the student team consisting of **Kiran K. Siddeshwar (01FE17BEC080), Varsha G. Radder (01FE17BEC213), Pratiksha U. Danti (01FE17BCS138) and Namrata A. Nyamagoudar (01FE17BCS248)**. The project report has been approved as it satisfies the requirements with respect to the mini project work prescribed by the university curriculum for B.E (V Semester) in the School of Electronics and Communications of KLE Technological University for the academic year 2019-2020.

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-The project team

ABSTRACT

We propose a framework for occlusion removal and inpainting of heritage site images. The project emphasizes on having occlusion free heritage site images, which aids in reconstructing the more realistic 3D model of heritage sites. The crowd-sourced images of Indian heritage sites and monuments are found to have occlusions. In order to get rid of these occlusions, we use Mask-Region based Convolutional Neural Network (M-RCNN) to achieve instance segmentation for occlusion removal and there is a network for inpainting considering the semantics of the image based on contextual attention and Generative Adversarial Networks (GAN) approach which can synthesize novel image structures in the corrupted areas. We make use of heritage data-set for our implementation. This approach mainly focuses on 3D reconstruction of heritage sites.This work can further be extended to make it compatible on mobile devices.

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Chapter 1

Introduction

In the domain of computer vision, occlusion removal and inpainting plays an important role. Inpainting is the process of reconstructing the deteriorated parts of images and videos. A network is proposed consisting of Mask-RCNN module that performs instance segmentation for occlusion removal. Instance segmentation is the task of detecting and delineating each distinct object of interest appearing in an image. Later there is an inpainting module based on generative adversarial network (GAN) based implementation for filling in the missing regions which can further be used for 3D reconstruction of heritage sites. The architecture proposed by Satoshi Iizuka et. al [3] consists of both global and local discriminator as adversarial losses which improved the inpainting result. Inpainting using Partial Convolution[4], where a binary mask is aided for convolving the holes with respect to the background pixel's data. Most of the current implementations work well for smaller patches but not for larger patches. In a nutshell, recent deep learning frameworks often generate blurry textures inconsistent with surrounding areas due to the ineffectiveness of CNNs.

1.1 Motivation

The heritage sites are getting affected by natural calamities and due to human intervention. Our project is intended to restore corrupted information digitally. To reconstruct the missing regions in a visually plausible manner such that it is reasonable to the human eye which is used for faithful reconstruction of 3D model. 3D model is constructed using the images, hence, the emphasis is lead to make images more realistic to enhance the 3d reconstruction.

1.2 Objectives

The main objective of this project is to remove the occlusion and inpaint the occluded images and train a inpainting model.

- Literature survey on methods for semantic segmentation inpainting.
- Design models for semantic segmentation and erasing unwanted areas in heritage images.
- Design models for inpainting of erased areas in heritage images.

- Demonstrate the results on real time heritage dataset towards 3D reconstruction.

1.3 Literature survey

Recent deep learning techniques have proved to be very robust in the field of object detection and localization. One such deep learning model is proposed by Kaiming He et. al. The Mask R-CNN[2] model is a simple, flexible framework for instance segmentation, which detects the occlusion and puts a mask around it. With some minor modifications, the model can generate a white mask and a binary mask image which is fed as an input to the proposed inpainting model.

A lot of image inpainting approaches have been proposed, which are either non-learning or learning-based. The earlier inpainting techniques used low-level features and completed the image based on conventional diffusion methods. Diffusion based means, the holes are filled using the neighborhood information of the original image. Such techniques work for images with smaller patches and with less structural and lighting information. For larger patches, this model yield blurry and smooth images. A similar approach is PatchMatch[1], which gave better results but it has a computational complexity that is more expensive and not practical for real-time applications. The limitation of these traditional methods is that the synthesized patch comes from the input image only, which is a constraint for the completion of the image that requires features that are not found in the input image.

The recent deep learning and GAN based approaches aim to fill larger missing regions and have given promising results. Pathak et. al proposed Context Encoder architecture[5] which semantically inpaints the image with standard pixel-wise reconstruction loss and adversarial loss, but the CE generates unrealistic or blurry images when missing regions have arbitrary shapes.

A different approach was used in inpainting using Partial Convolution[4], where a binary mask is aided for convolving the holes with respect to the background pixel data. This gave better result for smaller patches but not for larger patches.

In order to inpaint large missing patches, the other methods tend to use external data to predict the missing pixels. Whereas the Semantic inpainting using deep generative models[6] proposed by Raymond A, uses local method to predict the missing regions. The adversarial networks i.e. the generator and the discriminator are trained with good images (ground truth images). The similar encoded images are clustered in the latent space. When a corrupted image is given as input, in order to fill the missing regions the encoded image closest to the corrupted image is searched in the latent space. Here "closest" is defined by a weighted context loss to condition on the corrupted image, and a prior-loss to penalizes unrealistic images. The z vector is defined to find the closest cluster of features in the encoded images in the latent space and then z vector finds the closest features resembling the corrupted image's missing region. The closest image found fills the missing regions in the corrupted image.

The constraints of the context encoder was addressed by the architecture proposed by Satoshi Iizuka et. al [3]. This architecture consists of both global and local discriminator as adversarial losses which improved the inpainting results. However, the receptive field is increased using channel-wise fully connected layers in context encoders, whereas the Iizuka et. al proposed dilated convolution for the same.

There have been many deep learning algorithms that are practically used for face inpainting and generalized, but none or very few have implemented for Heritage sites dataset, which is a breakthrough for the crowdsource project.

1.4 Problem statement

Design a deep learning model to remove the occluded areas in the images and perform 2D Inpainting of Heritage Site Images. The inpainted images are further used for 3D reconstruction to store the information digitally.

1.5 Organization of the report

In chapter 2 System design we discuss the various method to design the system and choose the best design among the proposed design. In chapter 3 Implementation details we discuss the detailed implementation of the final design and each block in the final design. In chapter 4 Optimization we discuss the various optimization techniques used in the final design. In chapter 5 Results and discussions we discuss the results for the implemented final design and discuss the results. In chapter 6 Conclusions and future scope we discuss the improvement that can be brought to the final design and the future scope of the project.

Chapter 2

System design

Ihis chapter we discuss the different possibilities of the system design. Here we discuss the alternate design or multiple solutions by choosing different methods to implement each block and then choose the best design and describe each block in detail of the final design.

2.1 Functional block diagram

Initially our network takes images with occlusions as input. Occlusions are detected and localized using instance segmentation which is further removed and inpainted resulting in the occlusion free image, free from occlusions.

2.2 Morphological chart

As shown in Table 2.1 we discuss the different design possibilities of implementing our system to meet the objectives defined.

Table 2.1: Morphological chart

Functions/Means	**Option 1**	**Option 2**	**Option 3**
Occlusion detection and localisation	E-Net	F-B subtraction	Mask RCNN
Inpainting	Partial Convolution	Semantic Inpainting	Contextual Attention Modelling

2.3 Design alternatives

Option 1: Foreground - Background Subtraction and Semantic Inpainting this technique aims to remove occlusions from the image using two similar objects. Using two images, where occlusions are present at different locations, we can eliminate the occlusion. The

Semantic inpainting using deep generative models uses local method to predict the missing regions. The adversarial networks i.e. the generator and the discriminator are trained with good images (ground truth images). The similar encoded images are clustered in the latent space. When a corrupted image is given as input, in order to fill the missing regions the encoded image closest to the corrupted image is searched in the latent space.
Option 2: For localizing the occlusion, we can use semantic segmentation which identifies object outlines at the pixel level. E-net is used for the same, which uses a bottle neck architecture for pixel wise segmentation. A different approach was used in inpainting using Partial Convolution, where a binary mask is aided for convolving the holes with respect to the background pixel data. This gave better result for smaller patches but not for larger patches.
Option 3: The initial stage of the proposed pipeline is to perform occlusion detection and localization. Hence, we go for instance segmentation which identifies object outlines at the pixel level. Mask RCNN is a two-stage framework, the first stage scans the image and generates proposals (areas likely to contain an occlusion). And the second stage classifies the proposals and generates bounding boxes. A network for inpainting based on generative adversarial networks is proposed. The input to this network is an image with occlusion, masked with white pixels along with a binary mask of the same which gives the information of the holes. Instead of using a channel-wise fully connected layer, dilated convolution for increasing the receptive field. To further stabilize the training, the two-stage coarse-refinement network is used.

2.4 Final design

We select Mask RCNN for occlusion detection and localization and contextual attention architecture for inpainting, based on its working and ease of implementation.

Chapter 3

Proposed framework for occlusion removal and inpainting

In this chapter a complete explanation of each and every block used in the final design is given. It includes the detailed explanation about the data set used and the implementation details.

3.1 Specifications and proposed framework

The network consist of three main stages: Occlusion detection and localization, Mask generation and Inpainting.

3.1.1 Object Localization

The initial stage of our proposed pipeline as shown in Fig. 3.1 is to perform occlusion detection and localization. Hence, we go for instance segmentation which identifies object outlines at the pixel level.Mask RCNN consist of two stages.In the first stage the network scans the image and detect the region where occlusion is present and in the second stage bounding box is generated. Resnet-101 backbone is used as the feature extractor. A region proposal network (RPN) is present which is intended to find the areas that are likely to contain the desired object.Outputs of RPN are anchor class and bounding box refinement. In the bounding box refinement stage, non-maximal suppression is performed in order to get appropriate anchors. This produces final regions of interest (ROI).

3.1.2 Mask Generation

For erasing the occlusion, we generate a white mask (also called hole). The holes are generated by filling the occluded region with white pixels. The coordinates of the curve are obtained from the ROI obtained from M-RCNN. The mask obtained is post-processed using dilation to enhance the mask quality so that occlusion is removed completely. From the same idea, we also generate a binary mask that goes as an input to the inpainting network.

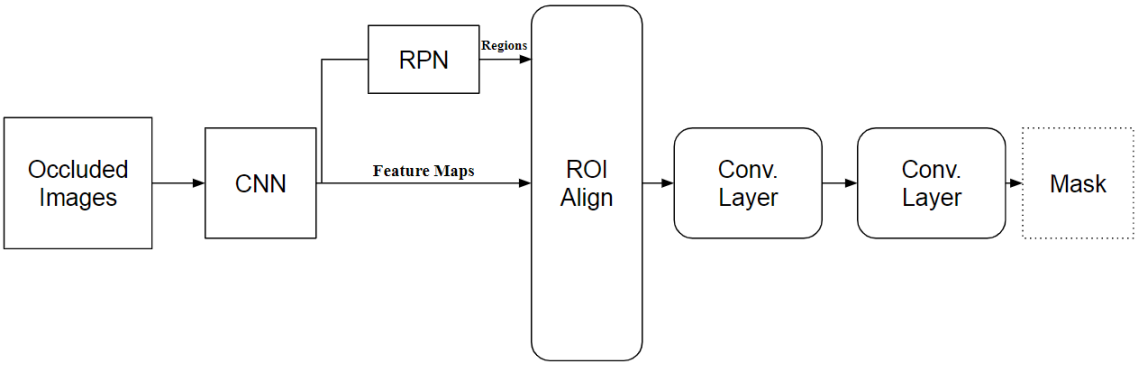

Figure 3.1: Block Diagram for object localization

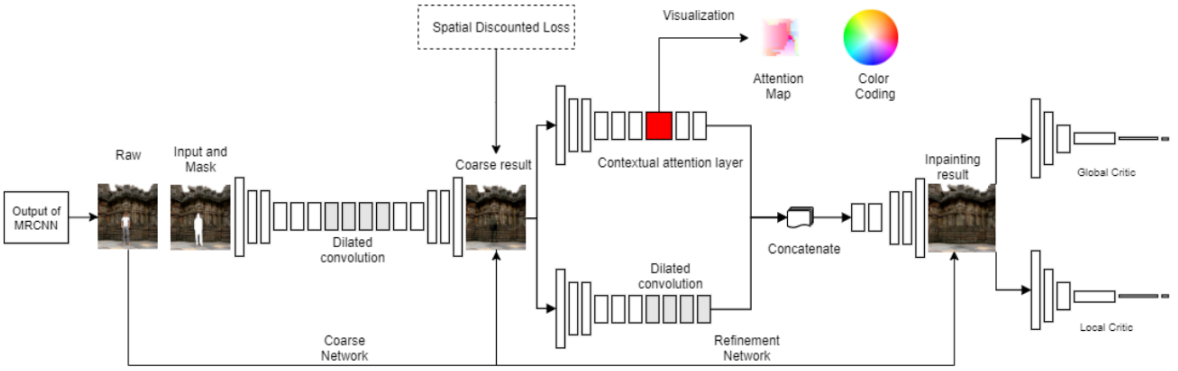

Figure 3.2: Inpainting Module

3.1.3 Inpainting module

We propose a network as in Fig. 3.2 for inpainting based on generative adversarial networks. The input to this network is an image of heritage sites with occlusion, masked with white pixels along with a binary mask of the same which gives the information of the holes. Instead of using a channel-wise fully connected layer, we use dilated convolution for increasing the receptive field. To further stabilize the training, the two-stage coarse-refinement network is used. The coarse network is trained for spatially discounted reconstruction loss. Missing pixels around the hole boundaries have less ambiguity than those pixels closer to the center of the hole. As the original image is used as the only ground truth for the computation of reconstruction loss, strong enforcement of reconstruction loss in those pixels may mislead the training process of the convolutional network. In this regard, spatially discounted reconstruction loss using weight mask M is introduced.

3.1.4 Refinement Network

The coarse result obtained is fed to two parallel encoders, one consisting of a contextual attention layer and other consisting of a dilated convolution layer, which is concatenated to a single decoder to obtain a refined output. The bottom encoder focuses on hallucinating pixels with dilated convolution. The contextual attention layer learns to facsimile feature information from the known patches to fill in the missing patches. A 3×3 patch from the background is extracted and reshaped to convolutional filters. The attention score of each pixel is propagated to get better patches, by adopting a left-right propagation followed by top-down propagation with a kernel size of k. Finally, deconvolution on attention score is performed to reconstruct foreground patches. The outputs of both encoders are concatenated and given to the decoder. The output image is fed to the local and global discriminator. These discriminators are used to check whether the image has been inpainted consistently or not. The global discriminator takes the full image as input and checks the global consistency of the image. The local discriminator takes the inpainted region and checks the local consistency of the image. The refinement network is trained with reconstruction loss and adversarial loss.

3.2 Algorithm

Algorithm 1 Training of our proposed framework.

while G has not converged **do**
  for $i = 1, ..., 5$ **do**
    Sample batch images $\mathbf{x}$ from training data;
    Generate random masks $\mathbf{m}$ for $\mathbf{x}$;
    Construct inputs $\mathbf{z} \leftarrow \mathbf{x} \odot \mathbf{m}$;
    Get predictions $\tilde{\mathbf{x}} \leftarrow \mathbf{z} + G(\mathbf{z}, \mathbf{m}) \odot (\mathbf{1} - \mathbf{m})$;
    Sample $t \sim U[0, 1]$ and $\hat{\mathbf{x}} \leftarrow (1 - t)\mathbf{x} + t\tilde{\mathbf{x}}$;
    Update two critics with $\mathbf{x}$, $\tilde{\mathbf{x}}$ and $\hat{\mathbf{x}}$;
  end for
  Sample batch images $\mathbf{x}$ from training data;
  Generate random masks $\mathbf{m}$ for $\mathbf{x}$;
  Update inpainting network G with spatial dis-
  counted ℓ_1 loss and two adversarial critic losses;
end while

Chapter 4

Optimization

4.1 Introduction to optimization

In this chapter we discuss about optimization, different types of optimization and selection of the suitable optimization technique. Optimization is a transformation technique which is used to reduce the amount of hardware resources and to improve the speed of the system.

4.2 Types of Optimization

The major types of optimization techniques used are:

1. Machine independent optimization: If the optimization technique does not alter the memory occupied during training or does not even reduce the training time then such optimization is called as machine independent optimization.

2. Machine dependent optimization: If the optimization technique used alter the memory required during training or reduces the training time and testing time then the optimization technique is called as machine dependent optimization.

4.3 Optimization Results:

In Fig.4.1, we compare the memory utilization of the code for images with different resolution i.e one image with original dimension and other with size of 512×512.

In table.5.1, we compare the time required for running the test code for images with different resolution i.e one image with original dimension and other with size of 512×512. The code profiling is as follows:

Optimization	**Before**	**After**
Time	27.279 secs	7.273 secs

Table 4.1: Time Comparison

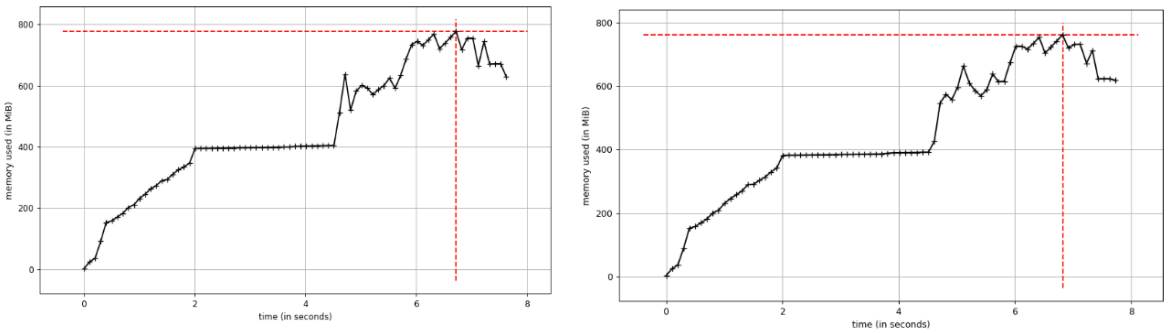

Figure 4.1: Memory Comparison

Although, python programming optimizes the code internally, other optimization technique includes usage of minimum loops to save time, modular approach to simplify the code, using functions and less global variables.

Chapter 5

Results and Discussions

In this chapter, we analyze and discuss the results obtained after training after training the network and then passing corrupted images through the generator.

5.1 Result Analysis and Discussion

The results of inpainting implementation starting with image with occlusion, binary mask and respective results of 11th epoch when we made use of DC-GAN. Each epoch was trained for more than 6 hours during the training process. The loss started to increase after 10th epoch, hence, we stop the training up to 10th epoch itself. We could infer that results were quite promising when we used DCGAN compared to when we used WGAN-GP.

Models	**PSNR**
DC GAN	34.63
WGAN-GP	31.56
Multicolumn GAN	11.9

Table 5.1: Quantitative Results

Figure 5.1: Result of WGAN-GP

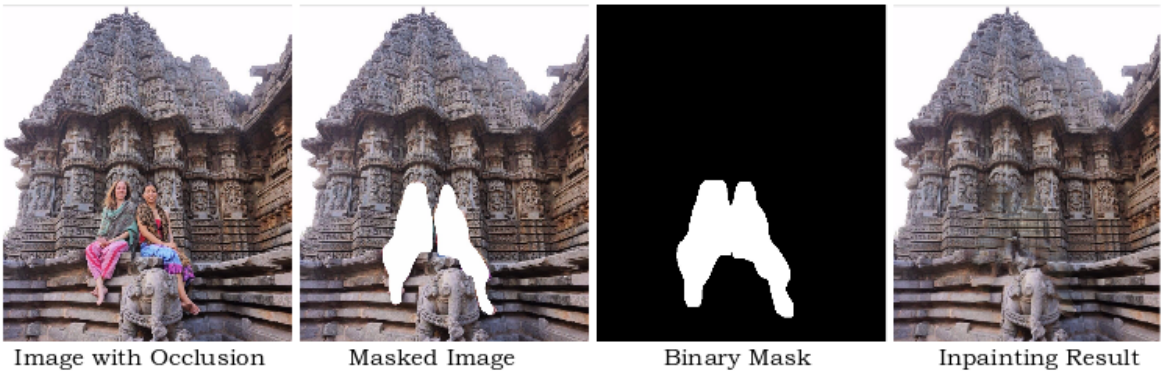

Figure 5.2: Result of DC-GAN

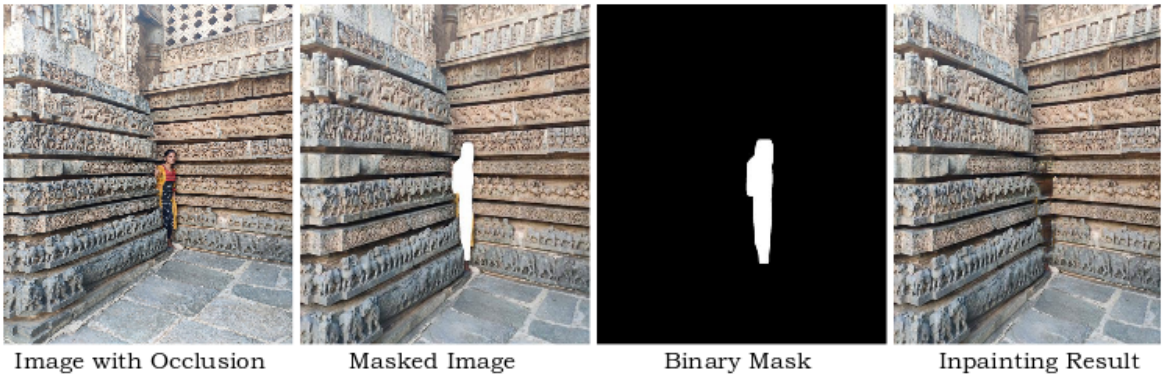

Figure 5.3: Result of WGAN-GP

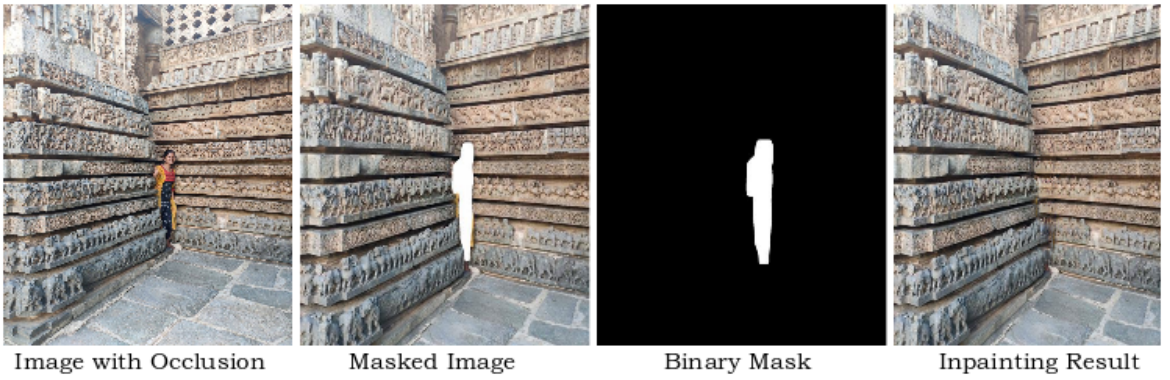

Figure 5.4: Result of DC-GAN

Chapter 6

Conclusions and future scope

In this chapter, the conclusions drawn and the future scope of the project are discussed following the implementation of the network and analysis of the result.

6.1 Conclusion

We propose a framework for occlusion localization and inpainting of heritage site images. Recent deep learning techniques have proved to be very robust in the field of object detection and localization. The paper emphasizes on having occlusion free heritage site images, which aids in reconstructing the more realistic 3D model. The use of a contextual attention module in our network has boosted the accuracy of inpainting. The use of DCGAN has improved the quality of inpainting. The results of this network indicate no requirement of post-processing of the inpainted images as they are at par from the other networks.

6.2 Future scope

As future work, we plan to extend to implement inpainting on mural paintings. The proposed inpainting framework can also be applied on conditional image generation and editing of the image. This framework can be further extended and modified towards underwater image inpainting. The proposed framework is applied to generate 3D models of several heritage sites.

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